

SOFT SWITCHING APPROACH TO REDUCING TRANSITION LOSSES IN AN ON/OFF HYDRAULIC VALVE

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ABSTRACT

A method for significantly reducing the losses associated with an on/off controlled hydraulic system is proposed. There has been a growing interest in the use of on/off valves to control hydraulic systems as a means of improving system efficiency. While on/off valves are efficient when they are fully open or fully closed, a significant amount of energy can be lost in throttling as the valve transitions between the two states. A soft switching approach is proposed as a method of eliminating the majority of these transition losses. The operating principle of soft switching is that fluid can temporarily flow through a check valve or into a small chamber while valve orifices are partially closed. The fluid can then flow out of the chamber once the valve has fully transitioned. Thus, fluid flows through the valve only when it is in its most efficient fully open state. A model of the system is derived and simulated, with results indicating that the soft switching approach can reduce transition and compressibility losses by 79%, and total system losses by 66%. Design equations are also derived. The soft switching approach has the potential to improve the efficiency of on/off controlled systems and is particularly important as switching frequencies are increased. The soft switching approach will also facilitate the use of slower on/off valves for effective on/off control; in simulation, a valve with soft switching matched the efficiency an on/off valve that was 5 times faster.

1 Introduction

Control using high-speed on/off valves has been proposed as a way to significantly improve the efficiency of hydraulic sys-

tems over the traditional method of metering valve control. Control using on/off valves, which mimics switch-mode converters in power electronics, saves energy by eliminating high pressure throttling of hydraulic oil through metering or relief valves. An example of an on/off controlled system is the virtual variable displacement pump proposed by Tomlinson and Burrows [1] and studied by Li et. al. [2], shown in Fig. 1. In this circuit, the on/off valve sends the full pump flow to either the load or the tank with minimal pressure drop, resulting in two highly efficient states. The output of the system is controlled through the use of pulse-width-modulation (PWM). This involves adjusting the amount of time that flow is sent to the load as a fraction of the fixed switching period. This fraction is defined as the duty ratio.

While on/off control has the potential for significant energy savings over metering valve control, several effects contribute to inefficiencies in a PWM controlled hydraulic system. Cao et. al. [3] suggest that energy can be lost in an on/off valve system as a result of valve transition times, fluid compressibility, pressure drops across the on/off valve(s), hysteresis in the accumulator bladder, fluid friction, and leakage. The authors have previously quantified a number of these sources of energy loss [4], [5] and found that, for the systems studied, the valve transition time was the most significant source of energy loss. Table 1 lists 4 mechanisms of energy loss and their relative magnitudes for the rotary on/off valve investigated in [5].

The valve transition loss arises from the fact that, as the on/off valve transitions between its two states, the open orifice area is decreased while oil is still flowing through it. This results in a high pressure drop across the valve and a significant amount of energy loss in throttling.

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Type of Loss	% of Total
Transition	60
Compressibility	20
Full Open Throttling	19
Leakage	1

Table 1. Relative Weights of Losses Estimated in a Rotary On/Off Valve Switching at 15Hz

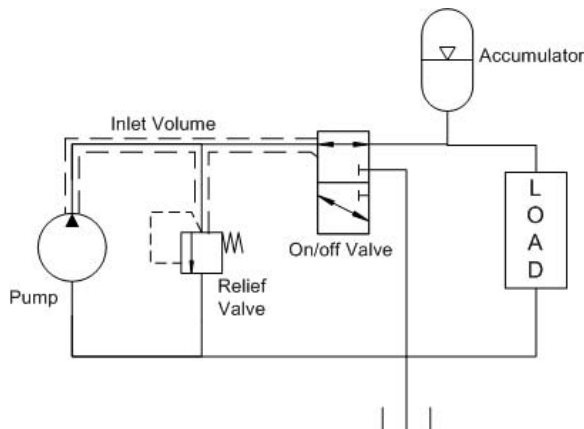


Figure 1. Virtual variable displacement pump using a 3-way on/off valve

A current focus of development in on/off controlled systems is to develop valves that can achieve high PWM frequencies, which are desirable for minimizing the system ripple and improving control bandwidth [4]. For efficient on/off control, the valve should spend as small of a fraction of the PWM period in transition as possible. However, as PWM frequencies are increased, this becomes increasingly hard to achieve. Since many valves are solenoid or pilot driven linear spool valves, the power required to accelerate and decelerate the valve spool increases proportionally to the switching speed cubed. Thus any reduction in transition losses by increasing the valve speed may be lost to valve actuation. In many valve designs, the transition loss per switch is fixed, and thus, as the PWM frequency is increased the loss due to transition throttling can grow significantly.

An alternative design that has been proposed by the authors is the unidirectional rotary on/off valve [5]. In this design, the valve spool rotates at a constant speed, and thus no power is required for spool acceleration. The actuation power will be dominated by viscous friction, which increases with the PWM frequency squared. In this design, the transition time decreases proportionally to the PWM frequency. Thus a parameter κ can be defined to represent the fraction of the PWM period that is spent during transition. For a given value of κ , the transition losses will be nearly constant with respect to the duty ratio [5].

In this paper, a new approach is proposed to significantly reduce the throttling loss during valve transition by providing

a temporary alternate path for the fluid when the valve area is small. This soft switching approach is similar to a technique in electronic switch-mode converters that uses small capacitors to temporarily absorb current when the transistors change state [6]. In an electronic system, soft switching is achieved by sizing components to induce large voltage and current ripples, which create temporary periods when the supply current flows in reverse. These large ripples are impractical in a hydraulic system. In this paper, we propose an alternative method in which an actively locked soft switch chamber is used in place of the traditional passive elements to avoid these large ripples.

The goal of applying soft switching to a hydraulic system is to reduce the transition loss so that higher PWM frequencies or higher κ values can be achieved efficiently. In addition, by reducing the transition time, the proposed soft switching approach can facilitate the use of much slower on/off valves. Since increasing the valve speed typically involves increasing the actuation power or decreasing the full open area, allowing slower valves can significantly reduce the energy lost in an on/off controlled system.

The basic concept of the soft switching approach is described in the next section, followed by a description of the system model in section 3. Section 4 explains simulations performed using the model, and section 5 presents a discussion of the results. Finally, concluding remarks are made in section 6.

2 Soft Switching Concept

While the soft switching concept can be applied to different system architectures, in this paper it will be described in conjunction with the three-way, virtually variable displacement pump configuration depicted in Fig. 1. When considering the operation of the three-way valve with soft switching, it is easier to imagine the three-way valve as two, two-way valves as shown in Fig. 2.

In a typical PWM cycle, beginning with the load valve fully open and the tank valve fully closed, the valve sequence is as follows: the tank valve will close during time interval t_1 , followed by the tank valve opening during t_2 . After some dwell time specified by the duty ratio, the tank valve will close during t_3 , which is followed by the load valve opening again over interval t_4 . Thus, in each PWM cycle there are 4 transition events during which throttling can occur. Figure 3 depicts area profiles for the two valves following linear area trajectories, which is typical of a rotary on/off valve.

In the simplest circuit, shown in Fig. 1, throttling will occur during each of these transition events. Figure 4 shows the pressure in the inlet volume for this circuit. Notice that significant pressure spikes will occur between t_1 and t_2 , and again between t_3 and t_4 when both valves are closed. These pressure spikes will be limited only by the compressibility of the oil in the inlet volume, and the pressure setting of the relief valve.

Soft switching can be applied to the load valve by adding a check valve in parallel as shown in Fig. 2. With this addition, the transitions of the load valve will be eliminated by allowing

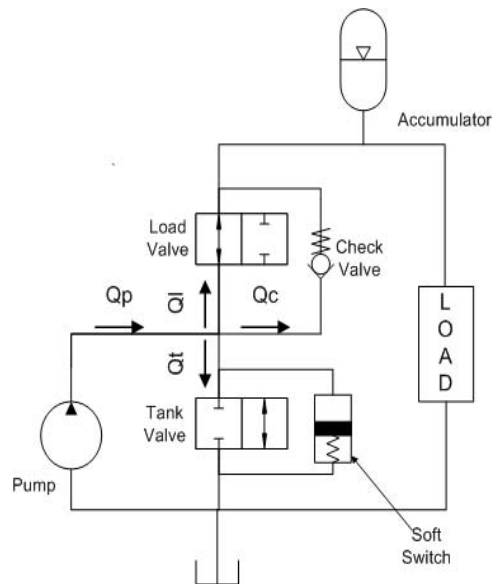


Figure 2. 3-way circuit with split tank and load valves, check valve, and soft switch

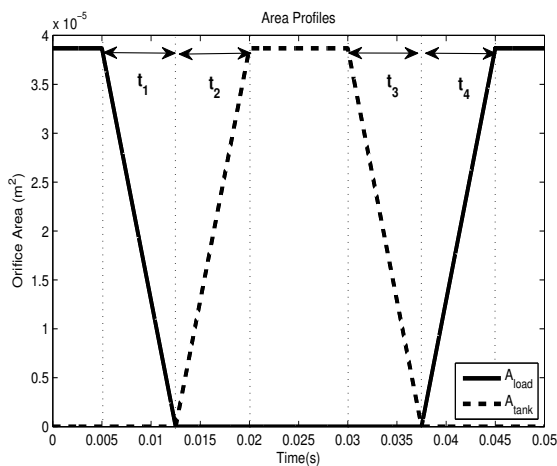


Figure 3. Area Profiles for 1 PWM Period

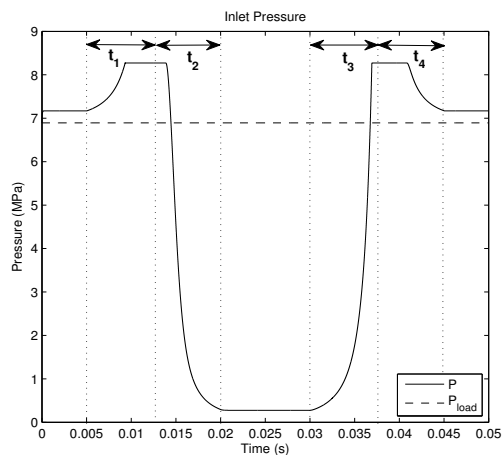


Figure 4. Inlet pressure profile for a 3-way circuit with a relief valve

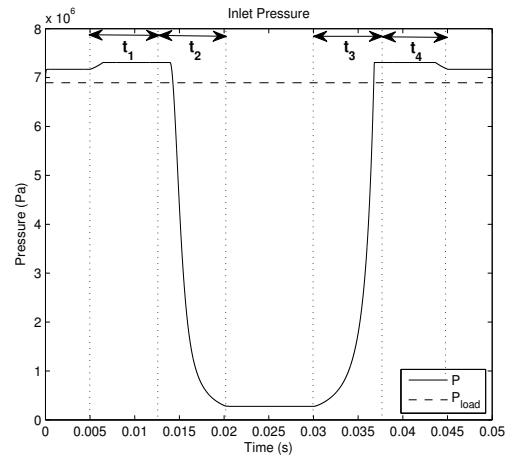


Figure 5. Inlet pressure profile for a 3-way circuit with a check valve across the load valve

oil to flow through the check valve rather than the partially open load valve. In some cases, the load valve could be completely replaced by the check valve.

The addition of a check valve dramatically reduces the energy loss due to throttling. However, a significant loss still exists as a result of the two remaining transitions (t_2 and t_3). Figure 5 shows the inlet pressure for a system with a check valve in parallel with the load valve and the area profiles shown in Fig. 3. The gradual transitions as the tank valve switches are evident. To eliminate these transition losses, the soft switch chamber shown in Fig. 6 can be used as a temporary storage mechanism.

The soft switch consists of a small chamber with a piston, a spring and an externally actuated locking mechanism. The locking mechanism holds the piston in roughly the middle of the chamber. Once the tank valve begins to open, the lock is released, and the high pressure fluid pushes the piston to the bottom of the chamber. Thus, the fluid is able to flow into the soft switch instead of being forced through the closing valve. Once the tank valve is fully open, the spring in the soft switch chamber will force the piston back up to the top of the chamber. Then, once the tank valve begins to close and the pressure rises above the spring pressure, fluid will once again flow into the soft switch. Once the piston hits the lock mechanism, the tank valve should be mostly closed, and the pressure will quickly rise and open the check valve. Finally, once the load valve is fully open, the check valve will close and the fluid will flow through the load valve.

3 Modeling

To study the operation of the soft switch, a model of the pressure dynamics and soft switch chamber is developed. In defining this model it is assumed that the output pressure is constant, the check valve opens instantaneously, there is no leakage, and fluid friction is negligible.

The area profiles for the transitioning valve were chosen to

be linear as shown in Fig. 3. While the area trajectory for a transitioning valve varies for different valve architectures, the linear case provides a simple example of the soft switch operating principle. In addition, the linear area trajectories represent the profile of a high-speed rotary PWM valve [5]. The soft switching principle will apply to any other type of valve trajectory, provided that the components are sized properly.

The pressure in the inlet volume of the circuit, which is depicted in Fig. 1, is given by the following equation:

$$\dot{P} = \frac{\beta(P)}{V_{in}} (Q_p - Q_t - Q_l - Q_c - a_s \dot{x}) \quad (1)$$

where V_{in} is the inlet volume, Q_p is the constant flow rate from the pump, Q_t , Q_l , Q_c are the flow rates through the tank, load, and check valves, a_s is the area of the soft switch piston, and x is the position of the soft switch piston. The bulk modulus of the hydraulic oil, $\beta(P)$, is modeled using the pressure dependent equation developed by Yu et. al. [7]:

$$\beta(P) = \frac{\beta_{oil}(1 + 10^{-5}P)^{(1+1/\gamma)}}{(1 + 10^{-5}P)^{(1+1/\gamma)} + 10^{-5}R(1 - c_1P)(\beta_{oil}/\gamma - 10^5 - P)} \quad (2)$$

where $\beta_{oil} = 1701 \text{ MPa}$ is the bulk modulus of air free oil, R is the ratio of entrained air volume to total volume, and $c_1 = -9.307 \times 10^6$ is a parameter describing the variation of air bubble volume due to changing amounts of dissolved air.

Flow rates through the tank and load valves are determined by the orifice equation:

$$Q = c_d a(t) \sqrt{\frac{2\Delta P}{\rho}} \quad (3)$$

where c_d is the orifice coefficient, $a(t)$ is the valve open area, and ρ is the density of the hydraulic oil. Note that the pressure drop across the fully open valve with full flow going through it is:

$$\Delta P_{open} = \frac{Q_p^2 \rho}{2a_0^2 c_d^2} \quad (4)$$

where a_0 is the full open area.

The check valve is assumed to open instantaneously at a pressure that is slightly higher than the full open pressure drop:

$$Q_c(t) = \begin{cases} 0 & \text{if } P < P_{load} + \Delta P_{check} \\ Q_p - Q_t - Q_l - a_s \dot{x} & \text{if } P \geq P_{load} + \Delta P_{check} \end{cases} \quad (5)$$

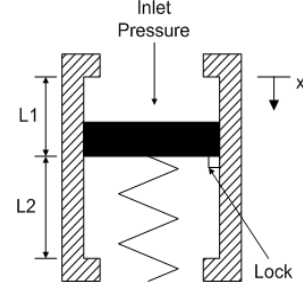


Figure 6. Diagram of the soft switch chamber

where $\Delta P_{check} > \Delta P_{open}$.

The soft switch chamber shown in Fig. 6 is modeled as a mass-spring-damper system with a pressure driving force:

$$\ddot{x} = \frac{1}{m_s} (Pa_s - F_s - b_s \dot{x}) \quad (6)$$

where m_s is the mass of the soft switch piston, b_s is the viscous damping coefficient, and F_s represents the spring force. For simplicity, it is assumed that F_s is a constant. This is a reasonable assumption if the initial spring force is large in comparison to the force variation. A linear spring can also be used without significant changes in the effectiveness of the soft switch.

In this paper, the piston lock is treated as an instantaneously acting mechanism that can be arbitrarily disengaged. There are several approaches for developing an actual locking mechanism, and a discussion of these can be found in section 5. The overall length of the soft switch chamber is split into two sections, l_1 and l_2 . When the piston lock is active, the piston is only allowed to travel between 0 and l_1 . Once the lock is disengaged, the piston can travel throughout the entire chamber.

4 Simulations

Using the equations derived in the previous section, a simulation of the three-way circuit with and without soft switching were conducted. The valve parameters were selected to approximate a three-way rotary on/off valve with a PWM frequency of 20Hz. The time for each transition was specified to be 7.5ms. The system is simulated with a flow rate of 37.9l/min and a load pressure of 6.89MPa. The fully open area was sized to have a pressure drop of 0.276MPa with $c_d = 0.65$ and $\rho = 876 \text{ kg/m}^3$. The inlet volume of the system was 30cm³, and the oil was assumed to have 3% entrained air. The check valve spring was set to be 0.138MPa. The system without a check valve or a soft switch chamber was simulated with a relief pressure 8.27MPa higher than the load pressure. In an actual system, this may need to be set much higher, depending on the required variation in the load pressure. As the relief pressure is increased, the losses in the system also increase.

The pressure profiles at 50% duty ratio for the relief valve

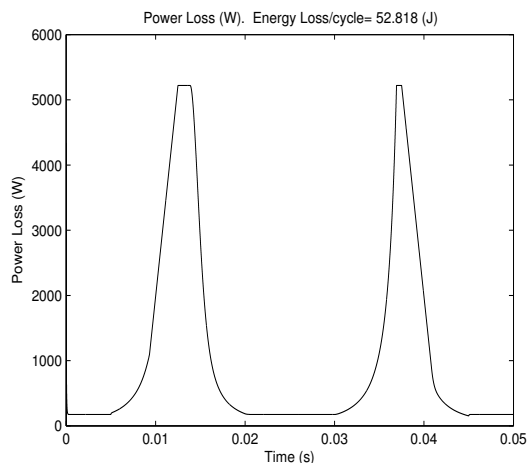


Figure 7. Power loss over 1 PWM period for a system with a relief valve

system and the system with a check valve across the load valve are shown in Figs. 4 and 5 respectively.

The power lost in the simulated system was calculated by subtracting the output power from the input power:

$$loss = PQ_p - P_{load}(Q_l + Q_c) \quad (7)$$

The power loss in the relief valve case is shown in Fig. 7, with a total loss of 52.8J for one cycle. In this figure, the large spikes in power loss due to the valve transition are evident. The goal of adding soft switching is to eliminate as much of these spikes as possible. Notice that the second loss spike, which occurs as the pressure is rising, is smaller than the first spike. This is due to oil compressibility; as the pressure drops, the oil expands and increases the total flow rate through the valve. The reverse occurs as the pressure is raised. Notice also the baseline loss of about 250W, which is due to the full open pressure drop. This accounts for about 8.7J of energy lost each cycle and cannot be addressed by soft switching.

By adding a check valve across the load valve, the power loss can be dramatically reduced. Figure 8 shows the power loss in the check valve system, with a total loss of 25.9J for one cycle. In this figure, the two large loss spikes have been reduced by about half, which is intuitive since the check valve is designed to eliminate losses in 2 of the 4 transition events.

To further improve the system efficiency, a soft switching chamber was added with a piston area of 1.2cm² and a piston mass of 12g. The first segment of the chamber had a length of $l_1 = 6mm$, and the second had a length of $l_2 = 30mm$. The constant spring force was set to be 91N, and the damping of the piston was set at 6Ns/m. The pressure profile for a system using these parameters is shown in Fig. 9.

Notice that the gradual transitions have been eliminated by the soft switch. The rapid movement of the soft switch piston as the pressure decreases results in a fair amount of pressure ring-

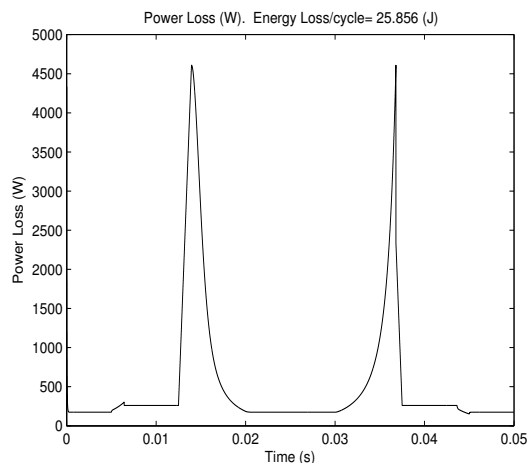


Figure 8. Power loss over 1 PWM period for a system with a check valve only

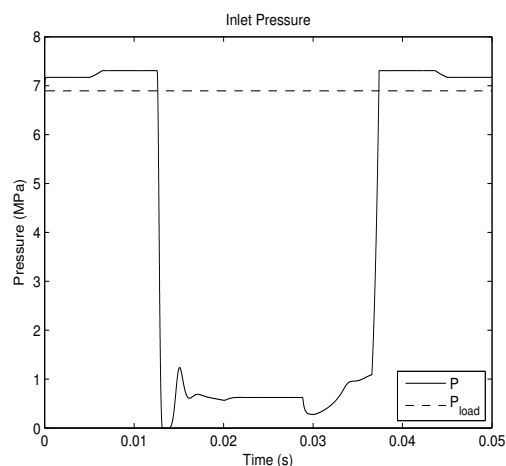


Figure 9. Pressure profile for a system with soft switching

ing. However, this may be reduced by applying damping in addition to the linear viscous damping (for example, a soft stop as is often used in hydraulic cylinders). The full open pressure between the two transitions is also significantly higher than in the case without soft switching. This is due to the emptying soft switch chamber, which maintains the higher pressure resulting from the spring force. Once the soft switch chamber is empty, the pressure drops down to ΔP_{open} before rising again as the chamber fills. This is the reason for the dip in pressure that occurs around 0.03s in Fig. 9.

The flow rates through the three different valves are shown in Fig. 10, which demonstrates the operating principle of the soft switch concept. During two portions of the cycle, the net flow through the three valves is less than the pump flow rate, and during another section, the flow is higher than the full pump flow. Thus, the soft switch chamber temporarily stores fluid while the valve is transitioning and then releases it once the valves are fully

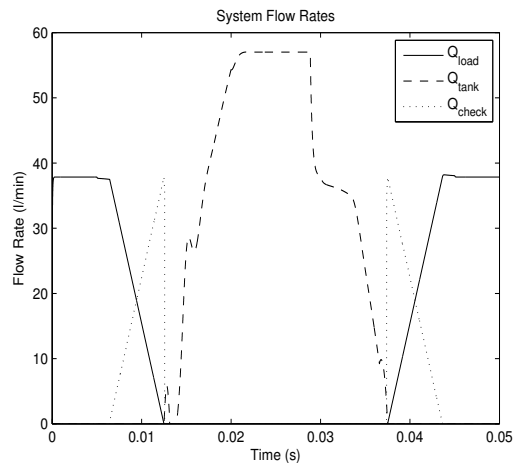


Figure 10. Flow rates in a soft switching system

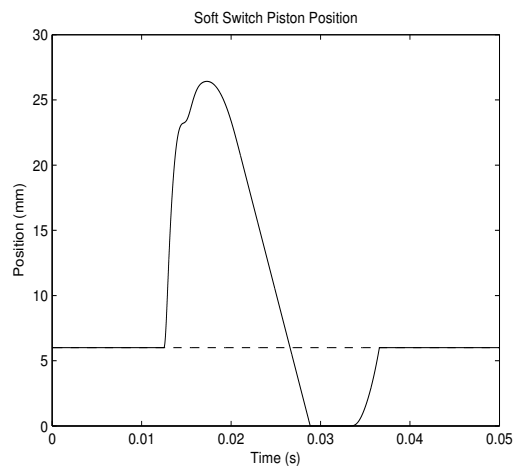


Figure 11. Position of the soft switch piston over 1 PWM period

open.

The position of the soft switch piston is shown in Fig. 11. At the start of the simulation, the piston is pressed against the locking mechanism. As soon as the lock is released, the piston leaves its initial position at l_1 . The piston position increases as long as the inlet pressure is higher than the spring pressure. Once the tank valve opens enough for the pressure to drop below the spring pressure, the soft switch chamber begins to empty. The spring force must be high enough for the piston to return past l_1 before the tank valve begins to close again. Thus the system components must be sized to ensure that the soft switch will operate properly over the desired range of operation. Another constraint on the system is that the inlet pressure must rise above load pressure before the load valve begins to open. Otherwise, high pressure fluid will flow back from the load, resulting in higher losses. This limits the length of l_1 .

The power loss over one cycle in the soft switching system is shown in Fig. 12. From this figure it is clear that the ma-

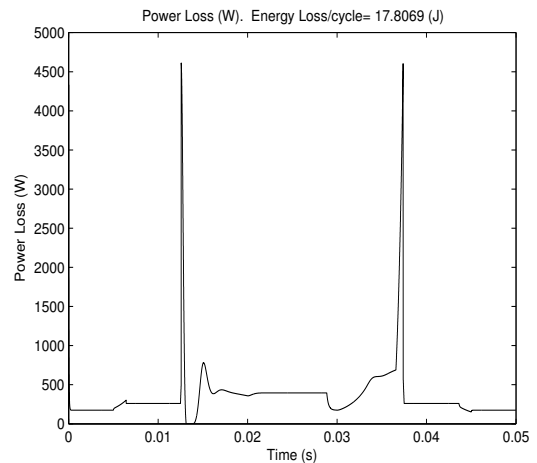


Figure 12. Power loss over 1 PWM period for a system with soft switching

System	Total Loss/cycle (J)
Relief	52.8
Check	25.9
Check and Soft Switch	17.8

Table 2. Loss per cycle for system with and without soft switching

jority of the large transition spikes have been eliminated by the soft switch. The remnant of the first spike is due to the fact that the soft switch piston has a finite acceleration and requires some energy to move. The second remnant, which is larger than the first, is due to fluid compressibility. Energy must be expended to compress the oil to the load pressure before the load valve opens. This loss is dependent on the amount of air entrained in the fluid and the size of the inlet volume. The power loss between the two spikes is higher than in the system without soft switching due to the emptying soft switch chamber. However, it is evident from the fact that the total energy lost in one cycle is reduced to 17.8J that this additional loss is less than the energy that has been saved during transition. The loss per cycle for the three different cases is summarized in Table 2.

5 Discussion

Table 2 shows that adding a check valve in parallel with the load valve reduced the overall system loss by about 51% from a system using only a relief valve to limit pressure spikes. The addition of a soft switch chamber eliminated an additional 15% of the losses, for a 66% reduction in the overall loss by applying the soft switching concept. Notice that the relief valve was set to be only 1.4MPa above the load pressure. In many systems, the relief pressure must be set significantly higher to allow for higher load pressures. This would increase the losses in the re-

lief valve system, making the soft switching approach even more beneficial.

The full open loss for the simulated system was about $8.7J$, which cannot be reduced by the soft switching approach. Thus, looking at only transition and compressibility losses, the relief, check and soft switch systems lose $44.1J$, $17.2J$, and $9.1J$ respectively. These numbers indicate that the soft switching approach can eliminate 79% of the transition and compressibility losses that exist in the base relief valve system. This indicates the potential of soft switching to significantly improve the efficiency of on/off valve controlled systems.

To determine whether the ideal check valve assumption was valid, a system with a dynamic check valve model was also simulated. The check valve was modeled as a piston with an area of $3cm^2$, a mass of $15g$, and a maximum opening distance of $0.5mm$. Using this model, the check valve closed in roughly $0.25ms$, and the loss in the soft switching system was increased by about $0.4J$. The system operation was similar to the case with the ideal check valve, with a some additional pressure ripples. The simpler model was used in section 4 to highlight the operation of the soft switch chamber. However, it was evident that a realistic check valve can be designed that is feasible for use in a soft switching system.

The simulations presented in this section were done with a duty ratio of 50%. With a properly timed locking mechanism, the soft switching approach is valid for any lower duty ratio. By decreasing the duty ratio, the amount of time that the load valve is open is decreased. However, by design, the soft switch piston is locked at l_1 any time the load valve is open, so no timing issues arise from decreasing the duty ratio. The magnitude of loss in the soft switching system is not dependent on when the events occur, and thus the loss is not duty ratio dependent (assuming proper operation).

As the duty ratio is increased, the time that the tank valve is open is decreased, with the open time decreasing to $0s$ at 100% duty ratio. During one soft switching cycle, the soft switch chamber must fill and then empty while the tank valve is open in order for the system to operate properly. The speed at which the chamber empties is a function of the spring force, which forces the stored fluid through the tank valve orifice. As the duty ratio is increased, there will be a point where there is not enough time for the chamber to fully empty before it must start filling again. Once this point is reached, the soft switching approach can no longer eliminate throttling during all 4 transitions. However, once proper operation is no longer possible, the locking mechanism can be disabled, and 3 of the 4 transition losses can still be eliminated (loss during t_2 will remain). Fortunately, this occurs at higher duty ratios, where system efficiency tends to be higher due to the increased system output.

To gain a greater range of feasible duty ratios, the system can be sized to empty faster. This is done by either storing less fluid (i.e. not completely eliminating the power loss during t_2) or using a stronger spring. Both of these options result in additional power loss. Thus a design choice must be made between

efficiency and range of operation based on the expected operating conditions. As the PWM frequency is increased, this timing problem will also occur. However, if the transition time is similarly scaled, then less fluid will have to be stored, and the timing will be similar to the slower system.

The system presented in this paper was operated at $20Hz$ PWM frequency, which was sized to be used with an existing on/off valve for future experimental validation. The concept is valid at higher frequencies with the caveat that the timing issues described in the previous paragraphs are taken into consideration. In addition, the compressibility of the oil in the inlet volume of the system can decrease the system effectiveness at higher PWM frequencies. This is a feature of all on/off controlled systems, with or without soft switching. Thus, a system with a sufficiently small inlet volume is a prerequisite for a properly functioning soft switch system. For example, in simulation the system was able to operate effectively at $80Hz$ with a $2.0ms$ transition time when the inlet volume was decreased to $8cm^3$.

While system compressibility and complex transition profiles can make simulation an attractive design tool, some design rules of thumb can be derived for sizing the spring force, F_s and the lower half of the piston travel, l_2 , which are the most critical timing elements. Since the goal of soft switching is to absorb the flow during transition, the bottom half soft switch volume can be designed as:

$$a_s l_2 = Q_p t_t \quad (8)$$

where t_t is the time for one transition. This slightly over estimates the volume since, as is clear in Fig. 10, the entire pump flow is not absorbed for the entire transition. However, it is a reasonable estimate. The spring force can then be estimated through the fact that the entire chamber and the full pump flow must pass through the fully open tank valve:

$$F_s = a_s \left(\left(\frac{\frac{a_s l_2}{(1-d)T - 2t_t} + Q_p}{c_d a_0 \sqrt{\frac{2}{\rho}}} \right)^2 - \Delta P_{open} \right) \quad (9)$$

where d is the maximum duty ratio for proper operation.

In simulation, the method is feasible for different load pressures; the most significant difference results from the compressibility of the oil. At higher load pressures, more fluid is needed to compress the oil, and more fluid is released as the fluid decompresses. To account for this change in the timing design equation (9), the equation for the oil compressibility (2) can be used in the following equation to determine the additional volume of fluid that must be discharged while the tank valve is open:

$$\Delta V_{compress} = V_{in} \int_{\Delta P_{open}}^{P_{load}} \frac{1}{\beta(P)} dP \quad (10)$$

For the simulation above, the system was able to eliminate the majority of the transition losses when the load pressure was tripled to 20.7MPa with a similar impact on system efficiency. However, a higher load pressure will increase the force required by the locking mechanism as well as the stress on different components and leakage.

The concept has been described without discussion of the locking mechanism to be used. Several options exist for creating a lock with the required actuation speed, such as piezo electric actuators or pressure triggered mechanisms. Ideally, the locking mechanism will be triggered by an external source with knowledge of the valve position. However, if this level of precision is not possible, a mechanism that is triggered by the pressure difference between the load and the inlet as the tank valve opens could also be used. This approach, while simpler, will result in some additional loss due to the time delay required for the tank valve to open enough for the pressure to fall below the load pressure. The locking mechanism that is selected must be capable of quickly releasing under a high load. For example, the locking force required by the system simulated in this paper is about 825N, which would require a robust mechanism. However, some adjustments can be made to the piston area and travel length to reduce this number.

The valve simulated in this paper had a transition time of 7.5ms. In order to match the energy loss of the soft switching system with a faster valve, the transition time needs to be increased to about 1.5ms for the relief valve case in simulation. Note that these are the times for one transition event, whereas typical valve specifications are often given for 2 events (i.e. load open - closed and then closed - tank open). This represents a five-fold increase in speed, which will typically correspond to a significant increase in price and actuation power for the valve. Thus the soft switching approach has the potential to enable on/off control using lower speed valves.

6 Conclusion

A method of significantly reducing the power lost in an on/off controlled hydraulic system has been introduced. The soft switching concept is a method for providing temporary alternative fluid paths around a transitioning valve, and thus avoiding the associated throttling losses. A check valve can be used to avoid two of the four transitions that occur in a three-way PWM controlled circuit; the remaining two transitions need an additional device to be reduced. The soft switching chamber, which consists of a small piston with a spring and a lock, can be used to reduce the overall system loss. Simulations were performed based on a PWM valve operating at 20Hz, and it was found that application of the soft switching concept resulted in an 79% re-

duction in transition losses, and a 66% reduction in overall system losses. This approach can make the efficient approach of on/off valve control even more attractive as an option for controlling hydraulic systems. In addition, soft switching can facilitate the use of slower valves for on/off control by significantly reducing the power loss resulting from slow transition times. In simulation, the soft switching system had the same efficiency as a system with a valve that was 5 times faster. This will allow simpler, lower power valves to achieve results similar to faster valves. The next steps in the development of the soft switching concept are the selection of a locking method, and the construction of an experimental prototype.

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